Coupled Atmospheric Temperature and Pressure Prediction using Hybrid RF-L

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EXECUTIVE ABSTRACT & SYSTEM SCOPE

Individual thesis investigating the physical coupling of ambient temperature and barometric pressure (QNH) at VCBI. Demonstrates that a hybrid Random Forest - LSTM pipeline outperforms isolated single-variable forecasting models, directly improving density altitude safety calculations.

1. Research Problem and Motivation

Temperature and atmospheric pressure directly dictate air density, which governs aircraft lift, engine thrust, and takeoff run distance.

Predicting temperature and pressure in isolation neglects the thermodynamic interdependence governed by daily solar irradiance and diurnal heating cycles.

2. Methodology: Hybrid RF-LSTM Architecture

The research implements a two-stage hybrid framework: Random Forest captures non-linear tabular feature interactions and lag attributes, while LSTM recurrent units learn multi-hour temporal dependencies and diurnal shifts.

Evaluated on historical VCBI records from the Department of Meteorology across both Northeast and Southwest monsoon seasons.

3. Experimental Findings & Contribution

" Baseline LSTM: R2 = 0.25 (limited sequence history generalization)

" Baseline Random Forest: R2 = 0.68, MAE = 1.14

" Proposed Hybrid RF-LSTM: R2 = 0.84, MAE = 0.86 (superior explained variance)

" Validated density altitude output accurately identifies high-risk midday takeoff conditions.